

Chirag Aparadh

413-472-4734 chiragaparadh767@gmail.com linkedin.com/in/chirag-aparadh github.com/git-chirag chirag-aparadh.vercel.app

Education

University of Massachusetts Amherst

Master of Science in Computer Science

Expected Graduation: May 2028

Amherst, MA

- **Coursework:** Neural Networks, Information Retrieval, Systems for Data Science, Machine Learning

Ramrao Adik Institute of Technology

2023

Bachelor of Technology in Computer Science and Engineering (CGPA: 9.51/10)

Mumbai University

- **Coursework:** Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Computer Architecture

Professional Experience

Oracle Financial Services Software

July 2023 – August 2026

Software Engineer (Associate Consultant)

Mumbai, MH

- Built and maintained core Java EE microservices powering HDFC Bank's net banking platform (2M+ daily active users), delivering 99.99% uptime through highly available clustered deployments.
- Implemented a BioCatch bot-detection workflow for merchant transactions such as Amazon and Flipkart, triggering OTP authentication on suspicious activity, strengthening fraud prevention for 150K daily transactions.
- Engineered event-driven integrations with Apache Kafka to publish acknowledgment messages to external systems such as Zerodha after fund transfers, enabling reliable, decoupled communication across banking services.
- Modernized a legacy banking workflow by translating UI-driven business logic into backend service APIs, performing end-to-end debugging across service and database layers and improving reliability through standardized data contracts and validation.

Projects

JiraRL | Python, FastAPI, OpenEnv, Qwen3, QLoRA, GRPO, Hugging Face

June 2026 – August 2026

- Built a stateful Jira ticket-queue simulator for training LLM agents to triage, assign, and resolve support tickets in priority order, with isolated episodes, dependency constraints, and executable rewards.
- Generated and validated 3,500 hint-free state-action examples from a rule-based expert via automated schema, data-leakage, and train/test isolation checks; published datasets and adapters to Hugging Face.
- Fine-tuned Qwen3-0.6B with 4-bit QLoRA; completed 60/60 held-out in-distribution episodes with every one of 500 actions advancing real work and zero invalid actions.
- Built environment-backed GRPO with checkpointing and promotion gates; diagnosed action collapse and zero-gradient saturation in a 20-step canary, stopping a 100-step run after 20% of the planned compute.

Multimodal Image Search System | FastAPI, PyTorch, CLIP, Qdrant, Redis, AWS

January 2025 – May 2025

- Developed a FastAPI service that ingests product-image CSVs (up to 1,000 rows × 10 images) and makes images searchable using natural-language queries via 512-dimensional CLIP embeddings.
- Indexed image vectors in Qdrant for text-to-image retrieval, visual similarity, and near-duplicate detection; validated in a smoke test ranking the target image first at 0.994 cosine similarity.
- Designed a fault-tolerant fan-out pipeline using Celery, with idempotent retries that recover from worker failures; containerized and deployed via Docker, AWS ECR, ECS, and S3.

Technical Skills

Languages: Java, Python, C, JavaScript, SQL

Backend & Distributed Systems: Java EE, FastAPI, REST APIs, Microservices, Apache Kafka, Redis, Celery

Cloud & DevOps: AWS (ECS, ECR, S3), Docker, Kubernetes, Jenkins, Linux

Databases: Oracle Database, MySQL, MongoDB, Qdrant

ML/AI: PyTorch, CLIP, Qwen3, QLoRA, GRPO, NumPy, Pandas, Scikit-learn, Hugging Face, OpenEnv

Achievements

- Top-800 national finalist of the Meta × Hugging Face OpenEnv Hackathon (31,000+ teams).
- Ranked 1108 globally in [LeetCode](#) Biweekly Contest 145 and earned the Knight badge (rating: 1866).

Publications

Exploring the Role of Blockchain in Enhancing Supply Chain

IEEE ICDSNS 2023

- First author; built an Ethereum DApp for farm-to-fork traceability without intermediaries.